

Homework 1

Create a new Vivado project and add the provided **hw1.vhd** module and **hw1_tb.vhd** testbench files. The module has three processes that should function as an 8-input AND gate each.

1. Perform a behavioral simulation using the given testbench (it should run for at least 50ns)
2. Analyze the waveforms to determine which of the processes correctly model an 8-input AND gate
3. Use the debugging techniques presented in the Xilinx Tutorial to determine where the errors are

Which processes show incorrect behavior? Explain why they behave the way they do.

4. Fix the errors so all three processes correctly function as 8-input AND gates
What needed to be done to fix the problem(s)?
5. Introduce a generic constant into the entity declaration that specifies the width of the a input
6. Rewrite the models so they use the generic constant and work for any input size
7. Write up answers to all questions and together with modified source code submit everything on myCourses under Assignments → Hw1